

```
#include <stdio.h>
int main()
{
    int num, originalNum;
    int divisor = 2;
    int largestPrime = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &num);
    originalNum = num;
    while (num > 1)
    {
        if (num % divisor == 0)
        {
            largestPrime = divisor;
            num = num / divisor;
        }
        else
        {
            divisor++;
        }
    }
    printf("The largest prime factor of %d is: %d\n", originalNum, largestPrime);
    return 0;
}
```